

ATS Resume Scanner Pro

AI-Powered Resume Analysis & Optimization Tool

Tech Stack

Python, Streamlit, Groq AI, pdfplumber, python-docx

Project Type: AI-based Resume Analysis Web Application

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1. Introduction

1.1 Project Overview

ATS Resume Scanner Pro is an AI-powered web application designed to help job seekers optimize their resumes for **Applicant Tracking Systems (ATS)**.

The system provides:

- Resume vs Job Description analysis
- ATS compatibility score (0–100)
- Keyword matching & missing keyword detection
- Skill gap analysis
- Formatting and structure feedback
- Actionable resume improvement recommendations
- Interactive AI chat for resume guidance

The project uses **Groq-hosted LLMs** for intelligent analysis and **Streamlit** for a clean, user-friendly interface.

2. Problem Statement

Many job applicants face issues such as:

- Resumes getting rejected by ATS before reaching recruiters
- Missing important keywords from job descriptions
- Poor resume formatting that ATS cannot parse
- No clear feedback on why resumes fail

There is a need for a **simple, AI-powered tool** that evaluates resumes and provides **clear improvement suggestions**.

3. Objectives

The key objectives of ATS Resume Scanner Pro are:

- Analyze resumes against job descriptions using AI
 - Generate an accurate ATS compatibility score
 - Identify missing keywords and skills
 - Provide actionable recommendations
 - Support common resume formats (PDF, DOCX, TXT)
 - Offer a beginner-friendly, web-based interface
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4. System Architecture

4.1 High-Level Architecture Flow

User → Streamlit Frontend → Groq AI API → Resume Analysis → Results Display

4.2 Architecture Explanation

Frontend (Streamlit)

- Resume upload (PDF, DOCX, TXT)
- Job description input
- Displays ATS score, analysis, and recommendations
- Interactive chat with AI assistant
- Downloadable analysis report

Backend / AI Layer

- Uses **Groq API** for LLM-based analysis
- Processes resume text and job description

- Generates structured ATS feedback
 - Extracts numerical ATS score programmatically
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5. Technology Stack

5.1 Frontend

- Streamlit
- HTML & CSS (custom styling)

5.2 Backend / AI

- Python
- Groq API
- LLaMA / Mixtral / Gemma models

5.3 File Processing

- pdfplumber (PDF extraction)
 - python-docx (DOCX extraction)
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6. Project Structure

```
ATS_Resume_Scanner/  
|— app.py  
|— requirements.txt  
|— README.md
```

7. Resume Analysis Logic

7.1 How It Works

1. User uploads resume
 2. Text is extracted based on file type
 3. User pastes job description
 4. AI analyzes:
 - Keyword overlap
 - Skill relevance
 - Experience alignment
 - Formatting & structure
 5. ATS score (0–100) is generated
 6. Feedback and recommendations are displayed
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7.2 ATS Score Calculation

- AI provides a numeric score based on:
 - Keyword matching percentage
 - Skill alignment
 - Experience relevance
 - ATS-friendly formatting

The score is extracted using pattern matching from the AI response.

8. AI Assistant Module

8.1 Features

- Resume-specific questions

- Improvement suggestions
- Skill highlighting guidance
- Career-oriented feedback

8.2 Model Support

- llama-3.1-70b-versatile
 - llama-3.1-8b-instant
 - mixtral-8x7b
 - gemma-7b-it
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9. Environment Configuration

Backend / App Configuration

GROQ_API_KEY=your_api_key_here

10. Installation & Setup

Step 1: Clone Repository

```
git clone https://github.com/your-username/ats-resume-scanner.git
cd ats-resume-scanner
```

Step 2: Install Dependencies

```
pip install -r requirements.txt
```

Step 3: Run Application

```
streamlit run app.py
```

11. Key Features

- ✓ AI-powered ATS scoring
 - ✓ Resume vs Job Description comparison
 - ✓ Keyword gap detection
 - ✓ Skill analysis
 - ✓ Formatting recommendations
 - ✓ Multi-model AI support
 - ✓ Interactive resume chat
 - ✓ Downloadable analysis report
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12. Future Enhancements

- Resume auto-rewrite feature
 - Multiple job description comparison
 - Resume section-wise scoring
 - PDF optimized resume export
 - User authentication & history
 - Job role-specific ATS templates
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13. Conclusion

ATS Resume Scanner Pro demonstrates how **AI-driven analysis** can help job seekers improve their resumes and increase ATS success rates.

The project focuses on:

- Simplicity
- Practical feedback
- Real-world job application challenges
- Beginner-friendly AI integration

This project highlights skills in **Python, Streamlit, AI APIs, and full application design**.

